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Hiring Committee
OCARINA Project
Inria
France

To the Hiring Committee,

I am writing to express my strong interest in the PhD position focusing on Optimal Control and Experimental Design within the OCARINA project. As an AI Engineer with a rigorous mathematical background, I am deeply fascinated by the application of optimal control to complex dynamical systems, specifically in neuroscience.

During my Master's thesis on the "Cardiovascular Medical Digital Twin," I successfully modeled highly complex physiological systems using advanced mathematical techniques. I developed a 0D Windkessel **PI_DeepONet** that utilizes continuous regularization via PyTorch Autograd to enforce cardiovascular fluid dynamics. This experience in integrating physical and biological constraints directly into deep neural networks perfectly aligns with the mathematical analysis required for modeling neuroscience applications and optimal control.

I am highly motivated to bring my expertise in deep learning, physics-informed architectures, and mathematical modeling to the OCARINA project to advance experimental design in neuroscience.

Sincerely,

Ismail Aissaoui